SAYOOJ.K.P

Mechanical Engineer | AI/LLM Evaluation | Generative AI & Automation | Industrial & Manufacturing | QA/QC | Prompt Engineering | Ex-BPCL



B. Tech Mechanical Engineering

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Mechanical Engineer with 6+ years of experience across industrial engineering, refinery operations, plant maintenance, QA/QC, technical documentation, manufacturing content validation, and AI/LLM evaluation. Experienced in applying Generative AI, prompt engineering and AI-assisted workflows for research, technical analysis, content validation and productivity. Combines mechanical engineering expertise with modern Generative AI tools, analytical problem-solving and technical communication to deliver accurate and reliable outcomes.

CAREER HIGHLIGHTS

- 6+ years of professional experience across Mechanical Engineering, Plant Maintenance, QA/QC, Technical Content Development, and AI Evaluation.
- AI Generalist & Manufacturing Domain Expert – Mercor, evaluating LLM outputs for reasoning, accuracy, instruction adherence and engineering feasibility.
- Generative AI & Prompt Engineering, with practical use of AI tools for research, analysis, content creation, productivity and AI-assisted workflows.
- BPCL Kochi Refinery experience in plant maintenance, shutdown/startup operations, SAP MM and multidisciplinary coordination.
- B.Tech Mechanical Engineering – First Class with Honours, CGPA 8.16, University Rank Holder.

PROFESSIONAL EXPERIENCE

AI Generalist Expert – Mercor

Contract | Remote | 2026

- Evaluated AI-generated responses for factual accuracy, reasoning quality, instruction adherence, contextual understanding, and communication.
- Designed and tested prompts for mechanical engineering and technical scenarios to evaluate AI reasoning, technical accuracy, and instruction following.
- Provided structured feedback to assess and improve LLM response quality.
- Applied structured prompting and modern Generative AI workflows to analyze, test and validate AI-generated outputs across technical and general-use scenarios.

Manufacturing Domain Expert – Mercor

Contract | Remote | 2026

- Evaluated AI-generated mechanical engineering and manufacturing responses for technical accuracy, industrial feasibility, and process reliability.
- Validated AI outputs against engineering principles and real-world industrial practices.
- Reviewed AI-generated content covering maintenance engineering, production systems, reliability, quality control, thermodynamics, and fluid mechanics.

Senior Subject Matter Expert - Mech, QA/QC Reviewer, ALT text - Innovalance Learning systems Pvt.Ltd

Sept 2022 – Sept 2025

- Performed technical quality checks, error analysis, and approval processes to ensure accuracy of mechanical engineering content.
- Reviewed and validated technical documentation and educational engineering content for accuracy, consistency, and quality standards.
- Produced accessibility-compliant ALT text descriptions for engineering diagrams, technical illustrations, and educational content to support inclusive digital learning.
- Applied mechanical engineering knowledge across thermodynamics, fluid mechanics, heat transfer, machine design, and related subjects.
- Used AI-enabled tools to support content validation and improve workflow efficiency and turnaround time.

Subject Matter Expert (SME) - Relevel by Unacademy

May 2022 – Aug 2022

- Delivered domain support for mechanical concepts, improving learner understanding and question resolution efficiency.
- Simplified technical concepts to improve learner experience and reduce repeat query volume by 32%.

Translation and Transcription (Robot training Malayalam) - Exonerate, Kochi

Freelancer - Jan 2022 – Apr 2022

- Translated and transcribed 6000+ Malayalam voice/text data entries with 99% accuracy, supporting AI/robotic voice-training model development.

Graduate Engineer Trainee - Bharat Petroleum Corporation Limited (BPCL), Kochi

Jan 2020 – Oct 2021

- Supported shutdown and startup activities of plant operations including gas turbines, boilers, cooling towers, and nitrogen plant operations.
- Participated in preventive and corrective maintenance activities for critical refinery equipment.
- Worked with SAP MM processes including material requisition and resource tracking.
- Assisted in shutdown planning and execution while maintaining strict safety and operational standards.
- Coordinated with multidisciplinary teams to ensure timely completion of maintenance activities.
- Assisted supervisors in supervising contractors, field safety compliance, and documentation.

Mechanical Engineer - Green Hub Engineering Contractors, Kochi

Dec 2017 – Nov 2019

- Supervised on-site mechanical work and contractor activities to maintain strict compliance with safety and equipment specifications**.**
- Monitored work progress to ensure adherence to specifications and safety requirements.

Subject Matter Expert - Chegg India PVT.LTD (Part Time/Remote)

Aug 2017 – Dec 2021

- Solved advanced mechanical engineering problems with high solution accuracy and on-time delivery.
- Leveraged analytical thinking and domain expertise to ensure precision, timely delivery, and conceptual clarity in solutions.

EDUCATION

B. Tech – Mechanical Engineering (First Class with Honours) – 2017

Nehru College of Engineering & Research Centre — University of Calicut

CGPA: 8.16 — University Rank Holder

CORE COMPETENCIES

Mechanical & Industrial:

• Plant Maintenance • Manufacturing • Reliability Engineering • Troubleshooting • Preventive Maintenance • Equipment Maintenance • Root Cause Analysis

AI & Generative AI:

• AI/LLM Evaluation • Generative AI • Prompt Engineering • AI Content Validation • AI-assisted Research • AI Automation • AI Agent Workflows

AI Tools

• ChatGPT • Claude • Gemini • Grok • NotebookLM • Gamma • Napkin AI • Julius AI • Genspark • Google AI Studio • Claude Code • Canva

Technical Content & Quality:

• Technical Documentation • Technical Content Review • QA/QC • Content Validation • Accessibility-Focused Content • ALT Text

Enterprise & Digital:

• SAP MM • Microsoft Excel • PowerPoint • Word • MATLAB • Python Basics • C/C++

Additional Tools:

• AutoCAD • SolidWorks • Cyber Security Basics

PROFESSIONAL STRENGTHS

• Analytical & Problem-Solving • Technical Communication • Attention to Detail • Cross-Functional Collaboration • Adaptability & Learning • Organization & Time Management

CERTIFICATIONS

• SAP MM (Material Management) • Project Management Essentials (PMEC) • Lean Six Sigma White Belt • Six Sigma Yellow Belt • Seven QC Tools • Cyber Security Professional • Computer Programming in C & C++ • Positive Intervention and Situational Awareness Induction • Introduction to Agents — Mercor Academy

AI LEARNING & WORKSHOPS

• Mercor Prompt Academy & Rubric Academy Fellowship • be10X AI Tools & Claude Workshop • be10X AI Tools & ChatGPT Workshop • Freedom with AI Masterclass

PROJECTS AND SEMINAR

- AI-Powered Predictive Maintenance & Failure Risk Analysis | Python, Scikit-learn – **Project**
- Engineering Document Quality Assurance — ChatGPT-Assisted Project - **GitHub Project**
- Design & Fabrication of Concentric Tube Heat Exchanger Using Nanofluids - **Project**
- Experimental Investigation of Ethyl-Esterified Rice Bran Petrol Fuel Blend - **Project**
- Hydrogen–Ethanol Blending as an Alternative Fuel in SI Engines - **Seminar**

LANGUAGES

- English • Malayalam • Hindi • Tamil

HONORS & ACHIEVEMENTS

- University Rank Holder – Calicut University • IPATE Qualified • Scholarship – EdukateAKid

TRAININGS

- Indian Railway Palakkad MEMU Shed - **Implant Training (1 Month)**
- Prakash Industries Limited, Bangalore - **Industrial Visit**